

A Comparative Analysis of Decision Tree and Neural Network Classifiers for Iris Flower Classification

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Abstract: Machine learning classification is widely used to categorize data across various real-world applications. Among the many supervised learning algorithms, Decision Trees and Neural Networks are widely used because of their different learning mechanisms. This study compares the performance of these two algorithms using the Iris flower dataset. Both models were implemented in Python using the scikit-learn library and evaluated using accuracy, precision, recall, F1-score, confusion matrices, and training time. The Decision Tree achieved perfect classification accuracy with minimal training time, while the Neural Network also demonstrated high predictive performance despite requiring longer training time and producing one misclassified sample. These findings suggest that Decision Trees are well suited for structured classification tasks requiring interpretability and efficiency, whereas Neural Networks remain suitable for more complex classification problems involving non-linear relationships.

Keywords: classification, Iris dataset, Decision Tree, Neural Network

1.INTRODUCTION

Machine learning is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed (Mitchell, 1997). One of the most common applications of machine learning is classification, where an algorithm learns from labelled data and assigns new observations to predefined categories based on their characteristics (Bishop, 2006). Classification algorithms are widely used in many real-world applications, including medical diagnosis, financial fraud detection, email spam filtering, image recognition, and agricultural monitoring. As machine learning continues to be adopted across various fields, selecting an appropriate classification algorithm has become increasingly important because different algorithms exhibit different strengths depending on the characteristics of the data and the problem being solved.

Among the numerous classification algorithms available today, Decision Trees and Neural Networks are two widely used supervised learning techniques that approach classification in fundamentally different ways. A Decision Tree classifies data by applying a sequence of logical rules that repeatedly divide the dataset into smaller subsets until a final prediction is reached (Quinlan, 1986). In contrast, a Neural Network learns patterns from data through interconnected layers of artificial neurons that adjust their internal weights during training to improve predictive performance (Rumelhart et al., 1986). Although both algorithms are designed to solve classification problems, they differ significantly in how they learn from data and generate predictions.

Decision Trees represent a traditional rule-based approach to classification, where each prediction follows a sequence of logical decisions from the root node to a leaf node. This structure allows the decision-making process to be traced step by step, making it possible to understand how a particular classification is produced. Neural Networks, on the other hand, represent a modern data-driven approach that learns complex relationships directly from training data through multiple interconnected layers. Unlike Decision Trees, the prediction process of a Neural Network is distributed across numerous weighted connections, making it considerably more difficult to trace the reasoning behind an individual prediction. Understanding these differences provides a meaningful basis for comparing the behavior and performance of both algorithms under the same experimental conditions.

To perform this comparison, this study uses the Iris dataset, one of the most widely recognized benchmark datasets in machine learning. The dataset was introduced by Fisher (1936) and consists of 150 flower samples representing three iris species: *Iris setosa*, *Iris versicolor*, and *Iris virginica*. Each sample contains four numerical features—sepal length, sepal width, petal length, and petal width—which are used to classify the flowers into their respective species. Due to its balanced class distribution, relatively small size, and well-documented characteristics, the Iris dataset has become a standard benchmark for evaluating and demonstrating supervised classification algorithms.

The objective of this study is to compare the performance of a Decision Tree classifier and a Neural Network classifier using the Iris dataset. Both models are implemented in Python using the scikit-learn machine learning library (Pedregosa et al., 2011). Their performance is evaluated using accuracy, precision, recall, F1-score, confusion matrices, and visual analysis of the trained models. In addition to predictive performance, this study also examines the differences in how each algorithm reaches its predictions, providing a comparative understanding of their characteristics, behavior, and suitability for small supervised classification problems.

2.RELATED WORK

Machine learning classification has been extensively studied, leading to the development of numerous supervised learning algorithms for solving classification problems. Among these algorithms, Decision Trees and Neural Networks remain two of the most widely used approaches because they employ fundamentally different learning strategies while addressing the same classification task. As machine learning continues to evolve, comparing these algorithms has become increasingly important for understanding their behavior, predictive performance, and suitability for different types of data.

One of the earliest large-scale comparisons of supervised learning algorithms was conducted by Caruana and Niculescu-Mizil (2006). Their study evaluated several widely used classifiers across multiple datasets and demonstrated that no single algorithm consistently outperformed the others. Instead, the performance of each algorithm depended largely on the characteristics of the dataset and the classification problem. These findings highlighted the importance of selecting classification algorithms based on the requirements of a specific task rather than assuming one approach is universally superior.

More recent studies have continued this line of research by using the Iris dataset as a benchmark for comparing classification algorithms. Due to its relatively small size, balanced class distribution, and well-documented characteristics, the dataset enables different models to be trained and evaluated efficiently under the same experimental conditions. Its completeness and simplicity reduce the influence of data quality issues, allowing researchers to focus on the behavior of the algorithms themselves. Consequently, the Iris dataset has remained one of the most widely used benchmark datasets for evaluating and comparing supervised classification models.

Building upon previous comparative studies, this research focuses on a direct comparison between a CART-based Decision Tree and a Neural Network using the Iris dataset. Both models are implemented under the same experimental conditions and evaluated using identical performance metrics to ensure a fair comparison. In addition to predictive performance, this study also considers how each algorithm produces its classifications, providing a broader understanding of their characteristics and suitability for supervised classification tasks.

3.THEORY

3.1 Decision Tree

A Decision Tree is a supervised machine learning algorithm that classifies data by recursively partitioning it into smaller subsets based on decision rules derived from the input features. Its tree-like structure resembles a flowchart, where each decision directs the data along a particular path until a final prediction is reached. A Decision Tree consists of four primary components: the root node, internal nodes, branches, and leaf nodes. The root node serves as the entry point of the tree, where the first decision is made. Subsequent decisions are performed by the internal nodes, which continue partitioning the data based on the selected features until a leaf node is reached. The resulting branches represent the possible outcomes of each decision, and the leaf nodes contain the final class predictions. This hierarchical structure allows the classification process to be followed step by step, making Decision Trees one of the most interpretable supervised learning algorithms (Quinlan, 1986).

In this study, the Decision Tree model is implemented using the Classification and Regression Trees (CART) algorithm, which is the default Decision Tree implementation provided by the scikit-learn library. Unlike some Decision Tree algorithms that allow multiple branches from a single node, CART constructs a binary tree by dividing the data into two subsets at each split. To determine the most appropriate split, the algorithm evaluates different candidate features using the Gini Impurity criterion. Gini Impurity measures the likelihood of incorrectly classifying a randomly selected sample if it were assigned according to the class distribution within a node, and lower impurity indicates a more homogeneous grouping of samples. By recursively selecting the split that produces the greatest reduction in impurity, CART gradually builds a tree that separates the training data into increasingly distinct classes (Breiman et al., 1984; Pedregosa et al., 2011).

Once the Decision Tree has been trained, a new sample is evaluated by traversing the tree from the root node to a leaf node. The decision rules encountered during traversal were established during training using the Gini Impurity criterion. At each node, the sample is evaluated against the corresponding decision rule and directed along the appropriate branch until a leaf node is reached, where the predicted class is assigned. Because these decision rules are explicitly defined during tree construction, the entire classification process can be traced from the final prediction back to the root node (Breiman et al., 1984).

3.2 Neural Network Theory

A Neural Network is a supervised machine learning algorithm inspired by the biological neural networks of the human brain. Rather than relying on predefined decision rules, it learns complex patterns directly from data by adjusting numerical parameters during training. A typical Neural Network consists of three main components: an input layer, one or more hidden layers, and an output layer. Each layer contains interconnected artificial neurons that receive, process, and transmit information to the subsequent layer. This layered architecture enables Neural Networks to model complex relationships between input features and output classes, making them well suited for a wide range of classification tasks (Bishop, 2006).

During training, information first passes from the input layer to the hidden layer in a process known as forward propagation. Each neuron in the hidden layer computes a weighted sum of its inputs, applies the Rectified Linear Unit (ReLU) activation function, and forwards the resulting output to the next layer. The ReLU activation function performs a non-linear transformation on the weighted inputs before passing them to the subsequent layer. The output layer applies the Softmax activation function to convert the final outputs into class probabilities, and the class with the highest probability is selected as the prediction. The prediction is compared with the actual class label to calculate the prediction error. This error is propagated backwards through the network using the backpropagation algorithm, enabling iterative weight updates (Rumelhart, Hinton, & Williams, 1986; Pedregosa et al., 2011).

Once training is complete, the learned weights and biases remain fixed while the network is used to classify new samples. Each sample is evaluated by passing it through the network using the same forward propagation process employed during training, producing an output that represents the model's prediction. Unlike Decision Trees, where predictions can be traced through explicit decision paths, Neural Networks generate predictions through numerous interconnected numerical computations distributed across multiple neurons. Consequently, the reasoning behind an individual prediction is generally less transparent, even though the network is capable of modelling highly complex relationships between input features and output classes (Bishop, 2006).

Equations

ReLU: $f(x) = \max(0, x)$

Softmax: $\sigma(z_i) = e^{z_i} / \sum_j e^{z_j}$

4.METHODOLOGY

4.1 Dataset

The dataset used in this study was the Iris dataset, which contains 150 samples of Iris flowers belonging to three species: Iris setosa, Iris versicolor, and Iris virginica. Each sample is represented by four numerical features: sepal length, sepal width, petal length, and petal width. These feature measurements serve as the input variables, while the flower species represents the target class to be predicted. The dataset was obtained using the `load_iris()` function provided by the scikit-learn library (Fisher, 1936; Pedregosa et al., 2011).

4.2 Experimental Setup

The experiment was implemented in Python using several open-source libraries. Scikit-learn was used for dataset loading, model implementation, and performance evaluation. Pandas was used to organize the dataset into a DataFrame, NumPy supported numerical computations, and Matplotlib was used for visualization. After loading the Iris dataset using `load_iris()`, the data were converted into a Pandas DataFrame with descriptive feature names and class labels to facilitate data inspection and analysis. The dataset was then divided into training and testing sets using an 80:20 split.

4.3 Decision Tree Implementation

The Decision Tree model was implemented using the `DecisionTreeClassifier` provided by the scikit-learn library. As discussed in the theoretical background, the classifier employs the Classification and Regression Trees (CART) algorithm to construct a decision tree from the training data. Following model training, the resulting decision tree was visualized and feature importance scores were extracted to support interpretation of the trained model (Pedregosa et al., 2011).

4.4 Neural Network Implementation

The Neural Network model was implemented using the `MLPClassifier` available in the scikit-learn library. The network architecture consisted of a single hidden layer containing ten neurons. The hidden layer employed the Rectified Linear Unit (ReLU) activation function to process weighted inputs before forwarding them to the output layer. The output layer internally applied the Softmax activation function to convert the network outputs into class probabilities, enabling prediction of the most probable Iris species. Unless otherwise specified, the remaining hyperparameters were retained at their default values provided by the scikit-learn implementation, with the exception of

max_iter, which was increased to 1000 to allow sufficient training iterations. (Pedregosa et al., 2011).

4.5 Evaluation Metrics

The performance of both classification models was evaluated using standard classification metrics provided by the scikit-learn library. Overall predictive performance was measured using accuracy, while the classification report was used to obtain precision, recall, F1-score, and support for each Iris species. In addition, confusion matrices were generated to visualize the number of correct and incorrect predictions made by each classifier across all classes. These evaluation metrics form the basis of the comparative analysis presented in the Results section (Pedregosa et al., 2011).

5.RESULTS

5.1 Decision Tree Results

The Decision Tree model was trained and evaluated using the methodology described in Section 4. The experimental results obtained from the trained model are presented below.

Accuracy

The Decision Tree model achieved an overall classification accuracy of 100% on the testing dataset.

Classification Report

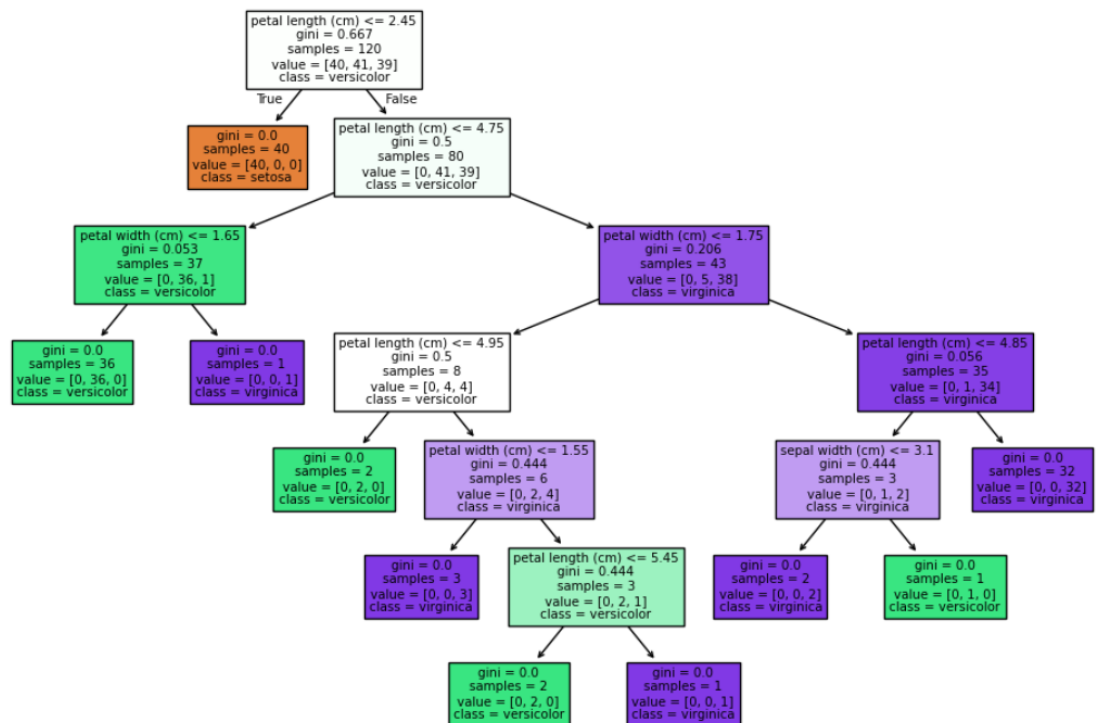
	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Confusion Matrix

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Decision Tree Visualization

The trained Decision Tree structure is presented below.



Feature Importance

The feature importance scores generated by the trained Decision Tree model are presented below.

```
sepal length (cm) : 0.000
sepal width (cm) : 0.017
petal length (cm) : 0.906
petal width (cm) : 0.077
```

Training Time

The Decision Tree model completed training in 0.0119 seconds.

5.2 Neural Network Results

The Neural Network model was trained and evaluated using the methodology described in Section 4. The experimental results obtained from the trained model are presented below.

Accuracy

The Neural Network model achieved an overall classification accuracy of 97% on the testing dataset.

Training Time

The Neural Network model completed training in 0.3764 seconds.

Classification Report

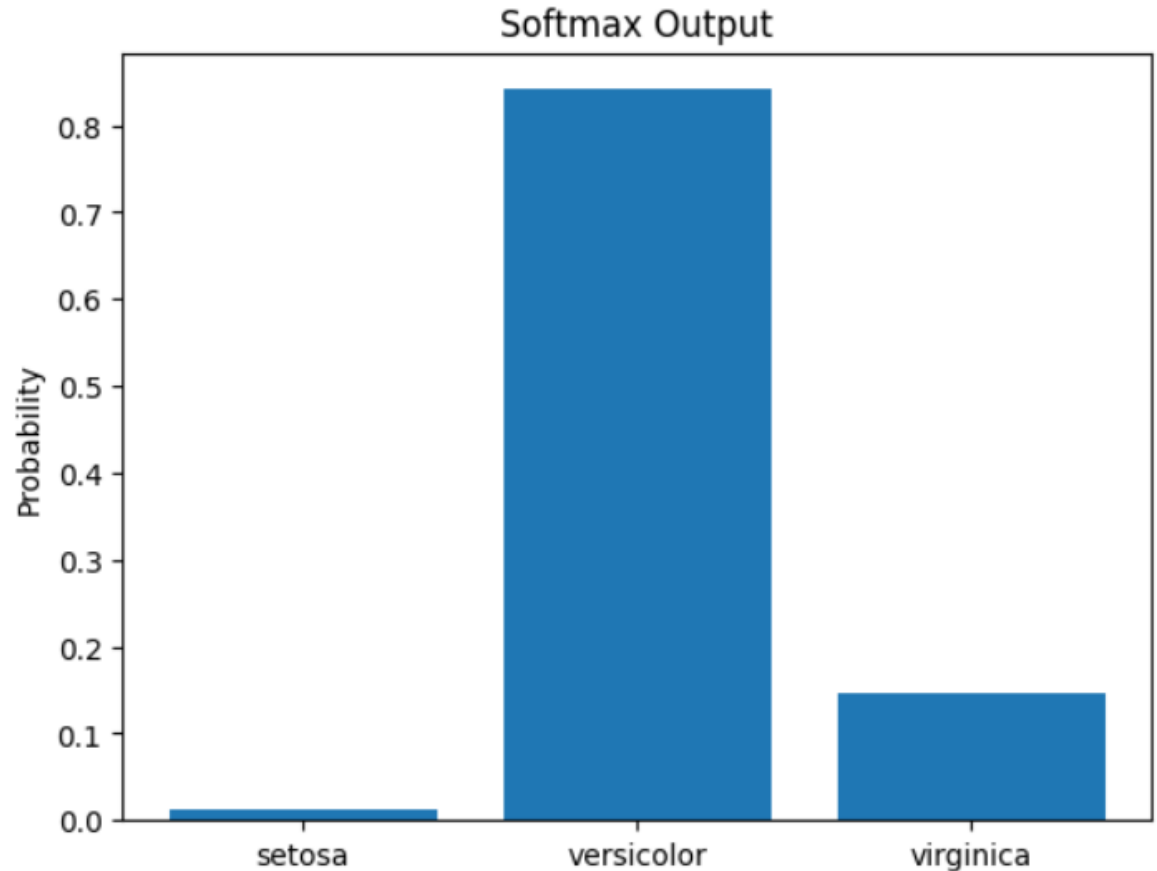
	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	0.89	0.94	9
virginica	0.92	1.00	0.96	11
accuracy			0.97	30
macro avg	0.97	0.96	0.97	30
weighted avg	0.97	0.97	0.97	30

Confusion Matrix

```
[[10  0  0]
 [ 0  8  1]
 [ 0  0 11]]
```

Predicted Probability Distribution

The predicted probability distribution produced by the Neural Network model is presented below.



6. Discussion

The experimental results showed that the Decision Tree classifier achieved perfect classification accuracy on the Iris dataset, whereas the Neural Network produced one misclassification, resulting in slightly lower performance across the accuracy, precision, recall, and F1-score metrics. This difference can largely be attributed to the characteristics of the Iris dataset, which contains relatively well-separated classes that are effectively separated by the recursive partitioning process of the CART algorithm. In contrast, the Neural Network learns decision boundaries by optimizing connection weights through backpropagation over multiple training epochs. Although this learning approach is highly effective for modeling complex non-linear

relationships, the relatively small and structured nature of the Iris dataset provided limited opportunity for the Neural Network to demonstrate this advantage. Consequently, the simpler Decision Tree model was able to achieve marginally better performance under the experimental conditions used in this study.

The differences observed between the two classifiers also reflect their underlying learning mechanisms. The Decision Tree model required considerably less training time because it constructed its classification rules through recursive data partitioning rather than iterative weight optimization. Furthermore, the Decision Tree was implemented using the default CART configuration without restricting the maximum tree depth or applying pruning techniques. While this approach resulted in perfect performance for the Iris dataset, unrestricted tree growth may increase the risk of overfitting when applied to larger or more complex datasets and therefore represents a limitation of the present implementation. In comparison, the Neural Network required approximately 31 times more training time due to repeated weight updates during training but maintained strong predictive performance despite its more complex learning process. Additionally, a convergence warning was generated during training, indicating the Neural Network had not fully converged within 1000 iterations and may not have reached its optimal weight configuration. This represents a limitation of the present implementation, analogous to the unrestricted tree growth limitation noted for the Decision Tree model. Overall, the observed differences demonstrate how algorithm design and dataset characteristics jointly influence classification performance, supporting the idea that different machine learning models are suited to different types of classification problems.

7. Conclusion

Based on the experimental findings, the Decision Tree classifier was highly effective for the Iris dataset, achieving perfect classification while requiring minimal training time. The Neural Network also demonstrated strong predictive performance, although it required longer training time and produced one misclassification. These findings suggest that Decision Trees are well suited for structured classification problems where interpretability and efficiency are priorities, whereas Neural Networks remain valuable for applications involving more complex data patterns and larger datasets.

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